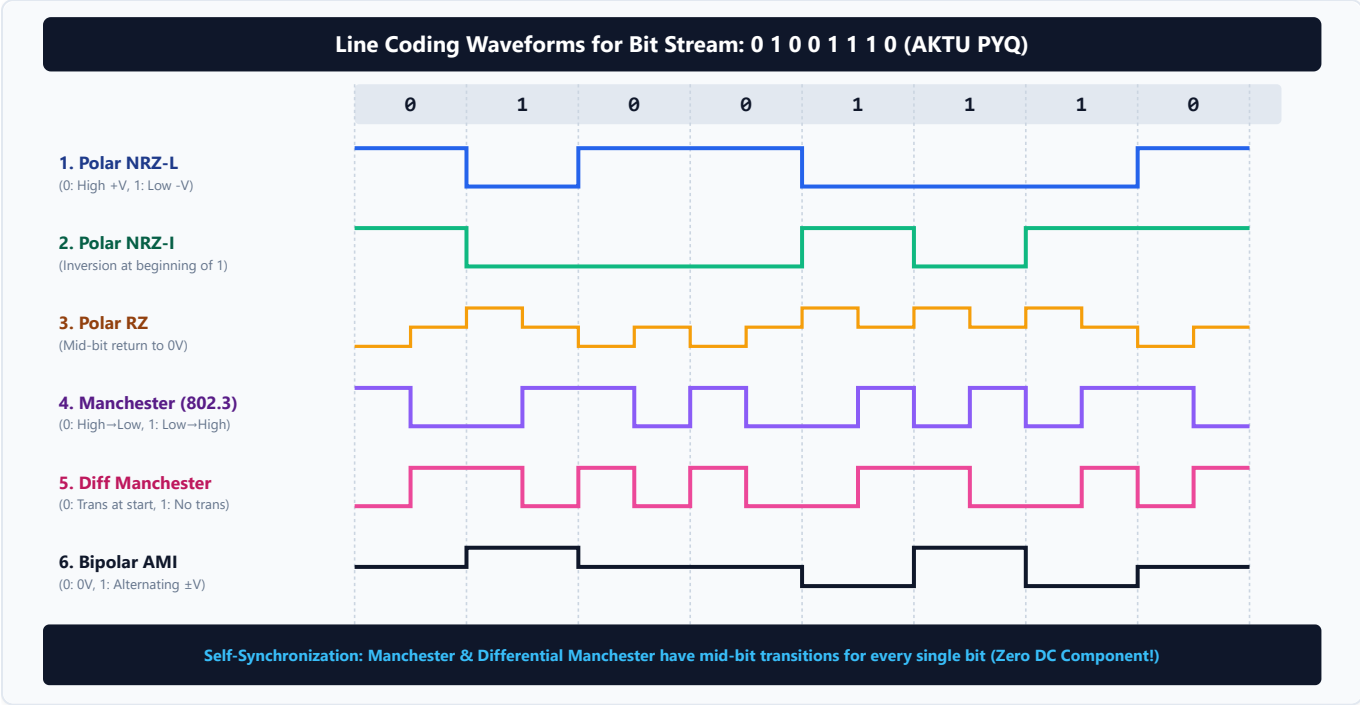


Module 11: Digital Transmission & Line Coding Techniques

AKTU BCS603 Exam Mandatory: Comprehensive waveform generation, transition rules, DC component elimination, and comparative analysis of Line Coding techniques.



1. Line Coding Schemes Comparison Matrix

Scheme	Baud Rate (S)	DC Component	Self-Sync Capability	Primary Practical Application
Unipolar NRZ	N / 2	Heavy (Undesirable)	None on consecutive bits	Obsolete legacy links
Polar NRZ-L	N / 2	Low to Moderate	None during long identical runs	RS-232 serial interfaces
Polar NRZ-I	N / 2	Low to Moderate	Syncs on 1s; loses sync on 0s	USB 2.0 (with bit stuffing)
Manchester	N (Double)	Zero (None)	Perfect (Mid-bit transition)	Classic 10Base-T Ethernet
Diff Manchester	N (Double)	Zero (None)	Perfect (Mid-bit transition)	Token Ring (IEEE 802.5)
Bipolar AMI	N / 2	Zero (None)	Syncs on 1s; loses sync on 0s	T1 / DS1 carrier telephone lines

AKTU Common Student Mistake: NRZ-I me transition sirf tab hota hai jab bit **1** aati hai! Agar bit 0 aati hai to signal bilkul change nahi hota. Is rule ko ulti mat draw karna!